Kinesiology & Biomechanics of Human Movement (Course: BPT-201)

Uploaded By: Rahul Joshi (BPT 4th Semester) | Official Course Curriculum | Peer Verified & 4.9

Unit 1: Kinematics & Kinetics of Human Joints

• Newton Laws applied to musculoskeletal movements: Force vectors, Center of Mass (CoM), Center of Pressure (CoP).

• Levers in the Human Body: 1st Class (Atlanto-occipital joint), 2nd Class (Plantar flexion of ankle), 3rd Class (Biceps brachii flexion).

• Convex-Concave Rule: When concave surface moves on convex, roll and glide occur in the SAME direction; when convex moves on concave, roll and glide occur in OPPOSITE directions.

Unit 2: Gait Cycle & Movement Pathologies

• Stance Phase (60%): Initial Contact (Heel strike), Loading Response, Mid-Stance, Terminal Stance, Pre-Swing (Toe-off).

• Swing Phase (40%): Initial Swing, Mid-Swing, Terminal Swing.

• Pathological Gaits: Trendelenburg Gait (Gluteus medius weakness), Hemiplegic Circumduction Gait, Parkinsonian Festinating Gait.